PYTHON FOR VIDEO GAMES WITH PYGAME · LESSON 2

The UFO moves with the keyboard

Goal: By the end you can draw a character with its Rect and move it with the arrow keys without leaving the window.
Before you start:

- Lesson 1 - your game.py with the window and the game loop.

To copy the code: when you copy from a PDF, many viewers drop the spaces at the start of each line, and in Python those spaces matter. Every step is also a `.py` file in the same folder as this guide: open it and copy from there.

Step by step

The UFO moves with the keyboard

In the last lesson you opened the game window; now the hero arrives: the UFO. First you load it, and learn how to know where it is. Then you move it with the arrow keys. And finally you stop it from escaping the window.

Step 2

Open your game.py. Below the background, load the UFO image: the file called player_1.png. convert_alpha keeps the transparent parts of the drawing, so you don't see a box around it. Now the important part: get_rect gives you a Rect, a rectangle that stores where the UFO is and how big it is. With midbottom you put the middle of its bottom edge at x four hundred, y five ninety; note: in pygame, y grows downward. And speed is how many pixels the UFO will move each frame. In the loop, after the background, you draw it with blit at that Rect's position.

Your `game.py` now looks like this — highlighted lines are new or changed (file `step-1.py`):

```
import pygame

pygame.init()
screen = pygame.display.set_mode((800, 600))
pygame.display.set_caption("Most Wanted UFO Mini")
clock = pygame.time.Clock()

background = pygame.image.load("assets/background_4.png").convert()
player_image = pygame.image.load("assets/player_1.png").convert_alpha()
player = player_image.get_rect(midbottom=(400, 590))
speed = 6

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
```

```
    screen.blit(background, (0, 0))
    screen.blit(player_image, player)
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

Step 3

Run the game. There is the UFO, at the bottom and in the middle. Notice the order: background first, then the UFO; what if you swapped them? You will find out in the quiz.

Try it:

```
python game.py
```

Step 4

Now, movement. Inside the loop, after the events, pygame.key.get_pressed() tells you which keys are held down right now. If the left arrow is held, you subtract the speed from player.x, its left edge, and the UFO moves left. If it is the right arrow, you add it. Since the loop runs sixty times per second, at speed six the UFO moves three hundred sixty pixels per second. Finally, clamp_ip pushes the Rect back inside the window, if it goes out.

Your `game.py` now looks like this — highlighted lines are new or changed (file `step-2.py`):

```
import pygame

pygame.init()
screen = pygame.display.set_mode((800, 600))
pygame.display.set_caption("Most Wanted UFO Mini")
clock = pygame.time.Clock()

background = pygame.image.load("assets/background_4.png").convert()
player_image = pygame.image.load("assets/player_1.png").convert_alpha()
player = player_image.get_rect(midbottom=(400, 590))
speed = 6

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        player.x -= speed
    if keys[pygame.K_RIGHT]:
        player.x += speed
    player.clamp_ip(screen.get_rect())

    screen.blit(background, (0, 0))
    screen.blit(player_image, player)
    pygame.display.flip()
    clock.tick(60)
```

```
pygame.quit()
```

Step 5

Run the game, and try the arrows. With the right arrow, the UFO goes to the edge, and stops there: that is clamp_ip at work. With the left arrow, it crosses the whole screen to the other side. Watch the panel on the right to see which key is held at each moment.

Try it:

```
python game.py
```

What you learned

Let's review. A Rect stores where an image is and how big it is. You draw the background first, and then the UFO. get_pressed tells you which keys are held, every frame. And clamp_ip keeps the UFO inside the window. Your exercise: make the UFO also move up and down with the up and down arrows. In the next lesson, the UFO shoots.

- A Rect stores position and size
- First the background then the UFO
- get_pressed reads the keys every frame
- clamp_ip keeps the UFO in the window

Summary

- `pygame.image.load(...).convert_alpha()` loads an image with transparency.
- `image.get_rect(midbottom=(400, 590))` creates a `Rect` the size of the image, placed by the middle of its bottom edge.
- `screen.blit(image, rect)` draws the image where the `Rect` is. Whatever you draw later goes on top.
- `pygame.key.get_pressed()` returns which keys are held this frame; `keys[pygame.K_LEFT]` is `True` while the left arrow is held.
- `rect.clamp_ip(screen.get_rect())` puts the `Rect` back inside the window.

Exercise

Start from this lesson's `game.py`.

- Make the UFO go up with the up arrow (`pygame.K_UP`) and down with the down arrow (`pygame.K_DOWN`).
- Check that `clamp_ip` also stops it at the top and bottom.
- Bonus: while `LSHIFT` is held, make the UFO move at double speed.

Solution

```
keys = pygame.key.get_pressed()
step = speed * 2 if keys[pygame.K_LSHIFT] else speed
if keys[pygame.K_LEFT]:
    player.x -= step
if keys[pygame.K_RIGHT]:
    player.x += step
if keys[pygame.K_UP]:
    player.y -= step
if keys[pygame.K_DOWN]:
    player.y += step
player.clamp_ip(screen.get_rect())
```

Art and sounds: Most Wanted UFO Classic, by Jose Daniel Soto, licensed CC BY 4.0.